Qualitative chemistry practice

1. Flame colours

State the characteristic flame colour of sodium, calcium and barium. Explain why a metal compound can give a characteristic colour.

2. Atomic orbitals

Name the three p orbitals. How many electrons can the complete p subshell hold?

3. Solubility product

Write the Ksp expression for AgCl(s). Explain why pure solid AgCl is omitted from the expression.

4. Inference

An unknown gives a strong yellow flame. Which element does this suggest? Give one reason to confirm the result by another test.

Answer key

1. Golden yellow, brick red, apple green. Electronic transitions emit light at characteristic wavelengths. 2. px, py, pz; six electrons. 3. Ksp = a(Ag+) x a(Cl-). The pure solid has unit activity. 4. Sodium; sodium contamination can dominate the flame colour.